

AI & MACHINE LEARNING SYSTEM DESIGN GUIDE

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SECTION 1: INTRODUCTION

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks requiring human intelligence.

This document provides a comprehensive overview of AI and ML concepts, system design approaches, best practices, and real-world applications.

SECTION 2: CORE DEFINITIONS

Term	Definition
Artificial Intelligence	Simulation of human intelligence in machines
Machine Learning	Learning patterns from data
Deep Learning	Multi-layer neural networks
Model	Mathematical function for prediction

Feature Input variable

Label Expected output

SECTION 3: TYPES OF MACHINE LEARNING

3.1 Supervised Learning

Supervised learning uses labeled datasets to train models.

Examples:

- Email spam detection
 - Credit scoring
 - Price prediction
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3.2 Unsupervised Learning

Unsupervised learning identifies patterns in unlabeled data.

Examples:

- Customer segmentation
 - Anomaly detection
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3.3 Reinforcement Learning

Reinforcement learning trains models using rewards and penalties.

Examples:

- Robotics
 - Game playing systems
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SECTION 4: ML SYSTEM PIPELINE

A typical ML system follows these steps:

1. Data Collection
 2. Data Cleaning
 3. Feature Engineering
 4. Model Training
 5. Model Evaluation
 6. Deployment
 7. Monitoring
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SECTION 5: OVERFITTING VS UNDERFITTING

Overfitting

- Model memorizes training data
- Performs poorly on unseen data

Underfitting

- Model fails to learn patterns
- Poor performance on all data

Best Practice:

- Use cross-validation
 - Apply regularization
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SECTION 6: REAL-WORLD APPLICATIONS

AI and ML are widely used across industries:

- Healthcare → Disease prediction
 - Finance → Fraud detection
 - E-commerce → Recommendation systems
 - Transportation → Autonomous vehicles
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SECTION 7: SYSTEM EXECUTION MODELS

Synchronous Processing

Synchronous systems wait for a response before proceeding.

Used for:

- Real-time inference
 - Payment validation
 - Authentication
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Asynchronous Processing

Asynchronous systems execute tasks in the background.

Used for:

- Model training pipelines
 - Logging and analytics
 - Batch predictions
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SECTION 8: MODERN BEST PRACTICES (2026)

Modern AI systems use a **hybrid execution model**:

Synchronous Use Cases

- Decision-making
- Real-time APIs

Asynchronous Use Cases

- Data pipelines
- Background jobs
- Monitoring

Key Recommendations:

- Minimize blocking calls
 - Use scalable architecture
 - Monitor system performance
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SECTION 9: LEGACY SYSTEM DESIGN (OUTDATED)

Older systems relied heavily on synchronous processing.

Issues:

- High latency
 - System bottlenecks
 - Poor scalability
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SECTION 10: TEMPORAL CHANGE IN BEST PRACTICES

Y Recommendation

2 Mostly synchronous
 systems

2 Mixed approach

2 Hybrid async-first
 architecture

SECTION 11: FAILURE HANDLING

Common system failures include:

- Network timeouts
- Data inconsistency
- Model drift

Solutions:

- Retry mechanisms
 - Monitoring tools
 - Fallback systems
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SECTION 12: LIMITATIONS OF AI SYSTEMS

- Requires large datasets
 - Can inherit bias from data
 - High computational cost
 - Lack of interpretability
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SECTION 13: CONFLICTING GUIDANCE (FOR TESTING)

Some experts recommend:

- Fully asynchronous systems for scalability

Others argue:

- Too much async leads to:
 - Debugging complexity
 - Loss of control
 - Execution order issues
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SECTION 14: EDGE CASES

Asynchronous Systems

- No guaranteed execution order
- Requires monitoring and retry mechanisms

Synchronous Systems

- Can block execution
 - High latency risk
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SECTION 15: LATEST TRENDS (2025–2026)

- Large Language Models (LLMs)
 - Generative AI
 - AI Agents
 - Autonomous systems
 - Explainable AI
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SECTION 16: UNRELATED INFORMATION (FOR TESTING)

Blockchain is a distributed ledger technology.

Quantum computing uses qubits instead of classical bits.

OAuth is an authorization framework for secure API access.

Data science vs. AI vs. ML

Data Science

- based on strict analytical evidence
- deals with structured & unstructured data
- includes various data operations



Artificial Intelligence

- imparts human intellect to machines
- uses logic and decision trees
- includes machine learning



Machine Learning

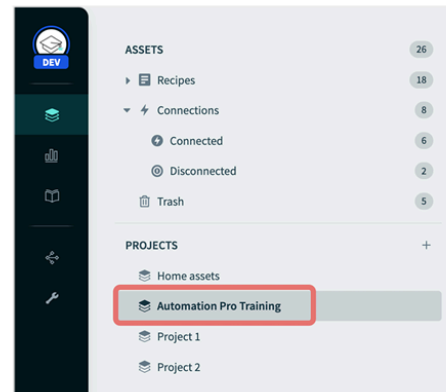
- subset of AI
- uses statistical models
- machines improve with experience



Lab 2: Create your first recipe

Start building your recipe

To begin building your recipe, navigate into the training project you created earlier.



Click the **Create** button.

You will see three options by default: recipe, connection, and folder. Depending on your account type, you may see more options.

For now, let's click **Recipe**.

This will begin the recipe building process.

